



Allo Nokia!!

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Difficulty: easy

Description

Before touchscreens and swipe keyboards, there was the legendary **keypad**. You'd tap the '**2**' **key thrice** just to get a '**c**', and sending a text felt like composing a symphony with your thumbs, especially with the sounds popping out when typing on a certain digit. In *Allo Nokia*, we're throwing it back to the good old days of snake games and polyphonic ringtones!





The mission is as simple as this:

1- **From Keypad to Message:** Given a sequence of key presses, return the message written.

2- **From Message to Keypad:** Now flip it. Take a normal text message and convert it into the exact sequence of button presses you'd use on an old-school mobile phone keypad.

Whether you were a pro at one-handed texting under the table or just love retro tech vibes, this challenge will test both your logic and your memory of the days when every letter was earned, one tap at a time.

Can you still text like it's 2005? Let's find out 📞

Inputs

- Two entries, the first entry is to decide if we will do the first of the second operation by giving "1" or "2".
- The second entry line is to enter the message we want to have the original typing integers for **OR** to enter the sequence of positive integers separated by spaces that we want to have the message for.
- Note that if the user typed "2222" for example it should loop back to "a" just like the old keypads.

Outputs

- The exact sequence of button presses you'd use on an old-school mobile phone keypad **OR** the original text respectively.

Constraints



- Itertools and libraries are prohibited and the teams who used it will have to resolve the challenge with a penalty!
- "0" is used for spaces.

Examples

Input	Output
1 666666 55 22 2 0 2222 555 555 2 666 88 2	okba allaoua
2 gdg algiers	4 3 4 0 2 555 4 444 33 777 7777
2 esi ex ini	33 7777 444 0 33 99 0 444 66 444
1 4 222 7 222 0 8 44 33 0 22 33 7777 8	gcpc the best